

Machine Learning

LAB EXERCISE 7

This lab exercise must be submitted by June 20th, 2025 at 10:00 am.
Late submissions will not be accepted and will be marked as zero.

This lab exercise is mandatory and could be assessed.
It is worth 1% of your total final grade for this course.

Submission Instructions

Submit your files through EClass. The files you submit cannot be read by any other students. You can replace your submission as many times as you like by resubmitting it, although only the last version sent is kept.

If you have last-minute problems with your EClass submission, email your assignment as an attachment to alberto.paccanaro@fgv.br with the subject "URGENT – LAB 7 SUBMISSION". In the body of the message, explain the reason for not submitting it through EClass.

All the work you submit should be solely your own work. Coursework submissions will be checked for this.
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In this lab, you will implement an unconstrained matrix factorization approach to recommend items to users. Note that we do not have any information about users and items other than the ratings users assigned to items.

DATASET DESCRIPTION

Movielens 100K Dataset: This dataset (in file “U.csv”) describes ratings ranging from 1 to 5 that users gave to movies. There are **943 users** and **1,682 movies**, and about **6.3%** of the user-item matrix entries are non-zero. This benchmark dataset is widely used to evaluate recommender systems.

YOUR TASKS

You will learn the latent factors for users and items using unconstrained matrix decomposition. This can be done by $R \sim UV^T$ where R contain the ratings, U contain user representations in its rows and V contain item representations in its rows. However, in this lab you will implement a more sophisticated approach.

Instead of modelling each rating as a dot product between user and item representations: $r_{i,j} \sim u_i v_j^T$, you will model the rating as $r_{i,j} \sim \mu + o_i + p_j + u_i v_j^T$. Where μ is the mean rating in the training set, o_i is the bias of user i , p_j is the bias of item j and the vectors u_i and v_j are the representations of the user and item, respectively.

You will build a function that learns the biases and a representation for each item and user from the known ratings in a training set using stochastic gradient descent. You should use weight decay to regularize your parameters during the learning. (Read page 108 of Aggarwal Book to better understand how to incorporate the biases into the learning and the gradient updates).

You will divide your dataset into train, validation and test sets. For each user, you will take out 10% of its movie ratings at random and split them: 5% for the validation set and 5% for the test set. Your training data matrix will have the same shape as the original matrix, but the ratings of the movies that were taken out should be set to 0.

For each user, you will predict the rating of the movies that were not rated in the training set. Then you will evaluate your predictions using the root mean square error (RMSE). The RMSE should be calculated between the known ratings in the validation/test set and the predictions made by your model. Additionally, you will compute the Hit Rate (HR) and the Average Reciprocal Hit-Rank (ARHR) at top-10 (you can read about this metrics in the SLIM paper available in e-class).

Today you should aim at finishing a few scripts that implement the above descriptions. During next week you should try to:

- make your code more modular, factoring it into functions;
- generate some plots (with title, variable names on the axes, etc).

Have fun ! ☺